AI-Assisted Software Development

Prompt Engineering for AI Assistants

Mike Burton

Why Do We Need Prompt Engineering?

Consider this scenario:

❌ "Write some code to handle dates"

- Vague
- No context
- No specific requirements

✅ "Create a function that parses dates in DD-MM-YYYY format and validates the input"

- Clear format
- Specific task
- Validation requirement

The difference? Prompt engineering! 🎯

The Mental Model

Think of AI as a brilliant but extremely literal intern:

- Knows a lot about coding
- Wants to help
- Needs clear, specific instructions
- Can't read your mind
- Will take your words exactly as written

This model helps us craft better prompts!

Key Principles

1. **Start General, Then Specify**
   - Begin with the big picture
   - Add specific requirements
   - Include constraints
2. **Provide Examples**
   - Show input/output pairs
   - Demonstrate edge cases
   - Illustrate error scenarios
3. **Break Down Complex Tasks**
   - Split into smaller parts
   - Build incrementally
   - Combine solutions
4. **Be Unambiguous**
   - Use precise language
   - State assumptions
   - Define terms clearly

Practical Example: Breaking Down Tasks

Instead of: ❌ "Generate a complete word search puzzle game"

Better approach: ✅ Break it down:

1. "Create a function to generate a 10x10 grid of random letters"
2. "Develop a function to place words into the grid"
3. "Write a function to display the grid"

This makes the AI's job easier and your results better! 🧩

Code Context Matters

When working with existing code:

```
def calculate_total(items):
# ... existing code ...
```

❌ "Make this better"

✅ "Refactor calculate_total to improve performance by using list comprehension instead of the current loop"

Context helps the AI understand your goals! 🎯

The Power of Examples

Consider this prompt:

```
Create a function to validate email addresses with these test cases:
Input -> Expected Output
"user@domain.com" -> true
"invalid.email" -> false
"user@domain" -> false
"@domain.com" -> false
```

Examples make the requirements crystal clear! ✨

Iterative Refinement

If at first you don't succeed... iterate!

1. Initial Prompt:

```
"Generate a date parser"
```

2. Better:

```
"Create a function that parses dates in DD-MM-YYYY format"
```

3. Even Better:

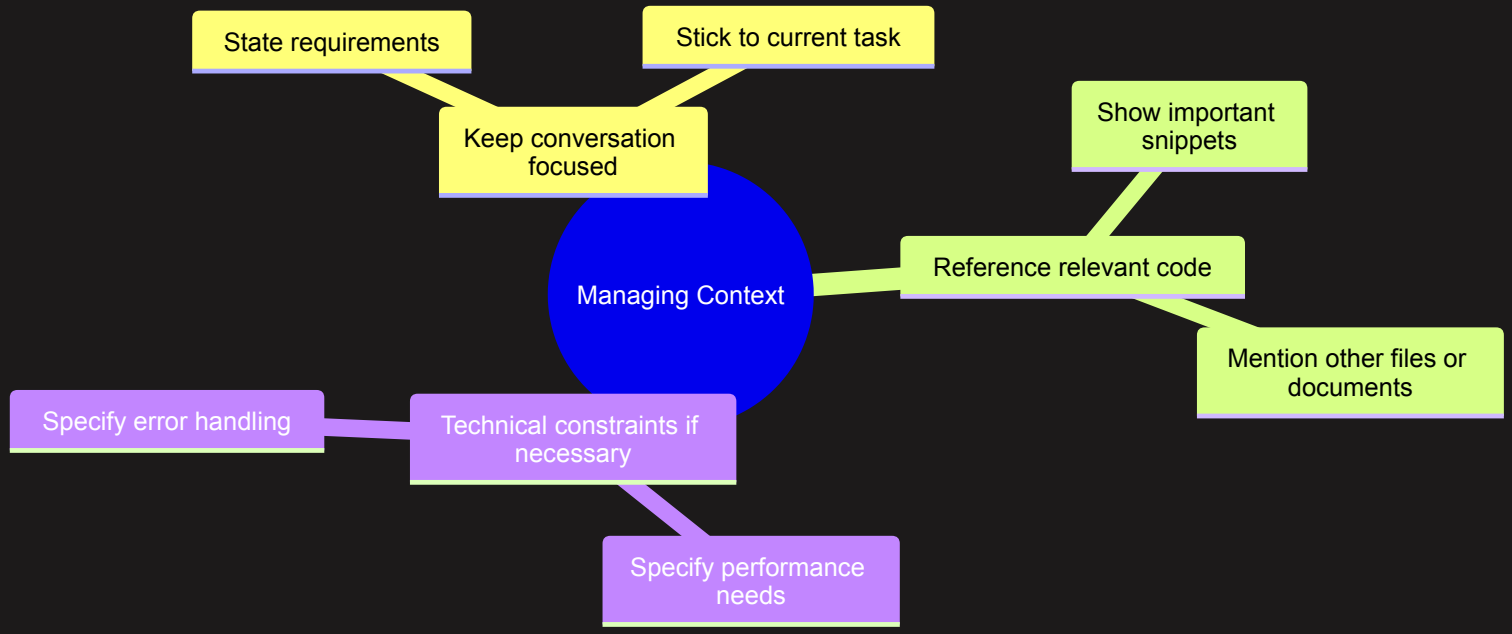
```
"Create a function that:
- Accepts strings in DD-MM-YYYY format
- Returns a DateTime object
- Throws an error for invalid dates"
```

Managing Context

When working with AI assistants, it's crucial to manage context throughout your interactions:

- keep the AI on track
- give it *enough* context
- but not too much (irrelevant information can decrease its accuracy!)

This ensures that the AI understands the current task and relevant details, leading to more accurate and helpful responses.



Best Practices

1. **Use Clear Variable Names**

```
# ✅ Good
user_score = calculate_average(scores)

# ❌ Avoid
x = calc(y)
```

2. **Specify Error Handling**
   - Describe expected exceptions
   - Define error messages
   - Outline recovery behavior

3. **Include Performance Requirements**
   - Time complexity goals
   - Memory constraints
   - Scalability needs

4. **Define Code Style**
   - Naming conventions
   - Documentation format
   - Testing requirements

Common Pitfalls to Avoid

1. **Being Too Vague**
   - ❌ "Make it better"
   - ✅ "Optimize the loop for better time complexity"
2. **Overcomplicating Prompts**
   - ❌ Long, rambling requirements
   - ✅ Clear, concise points
3. **Forgetting Edge Cases**
   - ❌ Happy path only
   - ✅ Include error scenarios

Summary

Key Takeaways:

1. **Be Clear and Specific**
    - Start general, then add details
    - Use precise language
    - Provide examples
2. **Break Down Complex Tasks**
    - Tackle one piece at a time
    - Build incrementally
    - Maintain context
3. **Iterate and Refine**
    - Start simple
    - Add detail as needed
    - Learn from results

Remember: Better prompts = Better code! 🚀